

No. of Printed Pages : 4
Roll No.

220942

4th Sem / Electrical
Subject : Electrical Machines - II

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 What is the frequency of DC?

- | | |
|----------|----------|
| a) 60 Hz | b) 0Hz |
| c) 50 Hz | d) 25 Hz |

Q.2 The Universal motor can work on

- | | |
|--------------|------------------|
| a) DC | b) AC |
| c) AC and DC | d) none of these |

Q.3 The Induction motor shaft is made of

- | | |
|---------------|--------------------|
| a) Mild steel | b) Stainless steel |
| c) Cast iron | d) All of these |

Q.4 The function of starter is

- | |
|-----------------------|
| a) To start the motor |
| b) To stop the motor |

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220942

- c) To start and stop the motor
- d) To limit the starting current

Q.5 The synchronous speed is given by

- a) $180f/P$
- b) $120f/P$
- c) $12f/P$
- d) $Pf/120$

Q.6 For ceiling fans, single phase motor used is

- a) Split phase motor
- b) Capacitor start motor
- c) Capacitor start and run motor
- d) Permanent capacitor motor

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Synchronous machine runs at _____ speed.

Q.8 LIM stands for _____

Q.9 Induction motor can rotate at Synchronous speed.(True/False)

Q.10 The value of coil span factor is always _____ than one.

Q.11 Slip of Induction motor is unit at stand still.(True/False)

Q.12 The Yoke of Induction motor is made up of _____

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Name different types of single phase induction motor and their uses.

Q.14 Draw and explain V-curve for Synchronous motor.

Q.15 Draw and explain Torque-slip curve of a Induction motor.

Q.16 Write a short note on split-phase motors.

Q.17 What do you mean by cogging and crawling in a three phase induction motor?

Q.18 What are different losses in Induction motor?

Q.19 Explain why stator core of an alternator is laminated?

Q.20 How can you reverse the direction of rotation of a three phase Induction Motor?

Q.21 Define Slip and its importance in a three phase Induction motor.

Q.22 Explain the working principle of Synchronous Generator.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain different types of speed control methods of Induction Motor.
- Q.24 Explain the construction and working of stepper motor with the help of diagram.
- Q.25 Explain the construction and working of Synchronous motor with the help of diagram.